

Mathematical Foundations of Political Methodology

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Day 6: Transforms

- Change of Coordinates
- Moment Generating Functions

Change of Coordinates

Proposition

Suppose X is a random variable and $Y = g(X)$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ that is a monotonic function.

Define $g^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ such that $g^{-1}(g(X)) = X$ and is differentiable. Then,

$$\begin{aligned} f_Y(y) &= f_X(g^{-1}(y)) \left| \frac{\partial g^{-1}(y)}{\partial y} \right| \text{ if } y = g(x) \text{ for some } x \\ &= 0 \text{ otherwise} \end{aligned}$$

Change of Coordinates

Proof.

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$$\begin{aligned}\frac{\partial F_Y(y)}{\partial y} &= \frac{\partial F_X(g^{-1}(y))}{\partial y} \\&= f_X(g^{-1}(y)) \frac{\partial g^{-1}(y)}{\partial y}\end{aligned}$$

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Then this is a pdf because $\frac{\partial g^{-1}(y)}{\partial y} > 0$.

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$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{\partial g^{-1}(Y)}{\partial y} \right|$$

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- Normal distribution
- Chi-Squared Distribution

Moment Generating Functions

Definition

Suppose X is a random variable with pdf f . Define,

$$E[X^n] = \int_{-\infty}^{\infty} x^n f(x) dx$$

We will call X^n the n^{th} moment of X

- By this definition $\text{var}(X) = \text{Second Moment} - \text{First Moment}^2$
- We are assuming that the integral converges

Moment Generating Functions

Proposition

Suppose X is a random variable with pdf $f(x)$. Call $M(t) = E[e^{tX}]$,

$$\begin{aligned} M(t) &= E[e^{tX}] \\ &= \int_{-\infty}^{\infty} e^{tx} f(x) dx \end{aligned}$$

We will call $M(t)$ the moment generating function, because:

$$\left. \frac{\partial^n M(t)}{\partial^n t} \right|_0 = E[X^n]$$

(Assuming that we can interchange derivative and integral)

Moment Generating Functions

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Differentiate once:

$$\begin{aligned}\frac{\partial M(t)}{\partial t} &= 0 + E[X] + \frac{2t}{2!}E[X^2] + \dots \\ M'(0) &= 0 + E[X] + 0 + 0 \dots\end{aligned}$$



Proof.

Differentiate n times



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Differentiate n times

$$\frac{\partial^n M(t)}{\partial^n t} = 0 + 0 + 0 + \dots + \frac{n \times n - 1 \times \dots 2 \times t^0 E[X^n]}{n!} + \frac{n! t E[X^{n+1}]}{(n+1)!} + \dots$$



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- If two random variables, X and Y have the same moment generating functions, then $F_X(x) = F_Y(y)$ for **almost all** x .

The Moments of the Normal Distribution

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$$M^6(0) = E[X^6] = e^{t^2/2}(t^6 + 15t^4 + 45t^2 + 15)|_0 = 15$$

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$$M^6(0) = E[X^6] = e^{t^2/2}(t^6 + 15t^4 + 45t^2 + 15)|_0 = 15$$

Proposition

Suppose X_i are a sequence of independent random variables. Define

$$Y = \sum_{i=1}^N X_i$$

Then

$$M_Y(t) = \prod_{i=1}^N M_{X_i}(t)$$

Proof.

$$M_Y(t) = E[e^{tY}]$$



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